1. Jelaskan Perbedaan antara routing static dan dynamic?

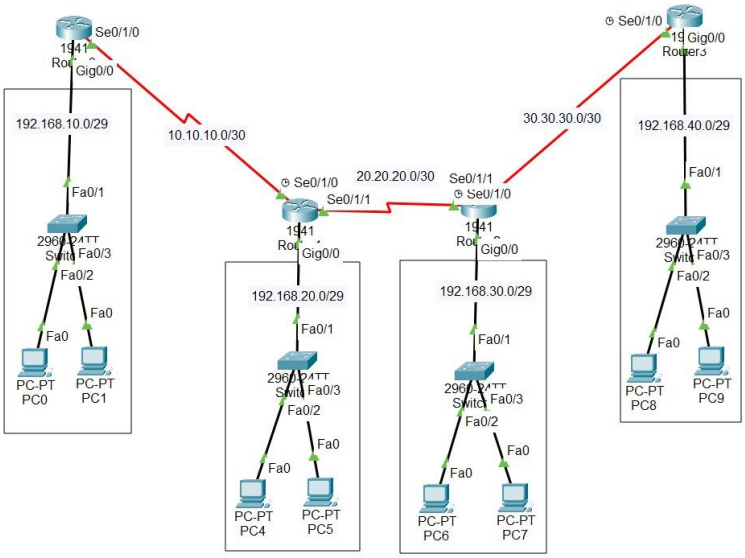
   Routing Statis adalah konfigurasi rute yang manual, bersifat tetap, dan cocok untuk jaringan kecil karena administrator harus memasukkan setiap jalur. Sebaliknya, Routing Dinamis adalah konfigurasi rute yang otomatis, dapat beradaptasidengan perubahan jaringan, dan ideal untuk jaringan besar karena router bertukar informasi rute menggunakan protokol *routing*.

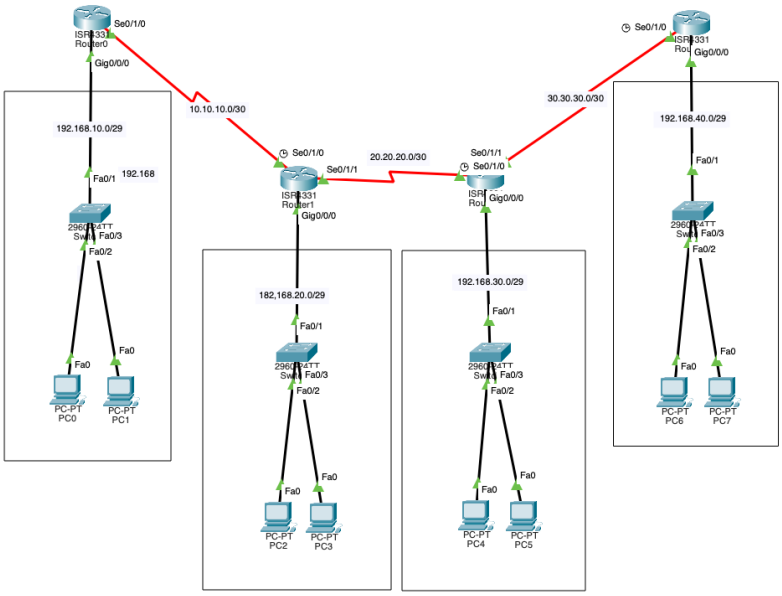
2. Jelaskan perbedaan antara Inderect Routing dan Direct Routing?

   Direct Routing (Perutean Langsung): Mengirim paket langsung ke tujuan karena keduanya satu jaringan fisik. Tidak butuh router.

   Indirect Routing (Perutean Tidak Langsung): Mengirim paket melalui satu atau lebih router karena tujuan berada di jaringan fisik yang berbeda. Wajib butuh router.

3. Buatlah rancangan topologi diatas pada perangkat masing masing, dan tuliskan configurasinya lalu tampilkan output hasil ping antar PC.





Router 0:

Router>enable

Router#conf t

Router(config)#hostname R0

R0(config)#int gig0/0/0

R0(config-if)#ip address 192.168.10.1 255.255.255.248

R0(config-if)#no shutdown

R0(config-if)#exit

R0(config)#int se0/1/0

R0(config-if)#ip address 10.10.10.1 255.255.255.252

R0(config-if)#no shutdown

R0(config)#ip route 192.168.20.0 255.255.255.248 10.10.10.2

R0(config)#ip route 192.168.30.0 255.255.255.248 10.10.10.2

R0(config)#ip route 192.168.40.0 255.255.255.248 10.10.10.2

Router 1:

Router>enable

Router#conf t

```
Router(config)#hostname  R1
R1(config)#int  gig0/0/0
R1(config-if)#ip  address  192.168.20.1   255.255.255.248
R1(config-if)#no  shutdown
R1(config-if)#int  se0/1/0
R1(config-if)#ip  address  10.10.10.2   255.255.255.252
R1(config-if)#no  shutdown
R1(config-if)#int  se0/1/1
R1(config-if)#ip  address  20.20.20.1   255.255.255.252
R1(config-if)#no  shutdown
R1(config)#ip  route  192.168.10.0   255.255.255.248   10.10.10.1
R1(config)#ip route  192.168.30.0   255.255.255.248   20.20.20.2
R1(config)#ip route  192.168.40.0   255.255.255.248   20.20.20.2
```

Router 2:

```
Router>enable
Router#conf  t
Router(config)#hostname  R2
R2(config)#int  gig0/0/0
R2(config-if)#ip  address  192.168.30.1   255.255.255.248
R2(config-if)#no  shutdown
R2(config-if)#int  se0/1/0
R2(config-if)#ip  address  20.20.20.2 255.255.255.252
R2(config-if)#no  shutdown
R2(config-if)#int  se0/1/1
R2(config-if)#ip  address  30.30.30.1   255.255.255.252
R2(config-if)#no  shutdown
R2(config)#ip  route  192.168.10.0   255.255.255.248   20.20.20.1
R2(config)#ip  route  192.168.20.0   255.255.255.248   20.20.20.1
R2(config)#ip  route  192.168.40.0   255.255.255.248   30.30.30.2
```

Router 3:

Router>enable

Router#conf t

Router(config)#hostname R3

R3(config)#int gig0/0/0

R3(config-if)#ip address 192.168.40.1 255.255.255.248

R3(config-if)#no shutdown

R3(config-if)#int se0/1/0

R3(config-if)#ip address 30.30.30.2 255.255.255.252

R3(config-if)#no shutdown

R3(config)#ip route 192.168.10.0 255.255.255.248 30.30.30.1

R3(config)#ip route 192.168.20.0 255.255.255.248 30.30.30.1

R3(config)#ip route 192.168.30.0 255.255.255.248 30.30.30.1

Ping Pengujian:

Ping PC0 ke PC2

c:\>ping 192.168.20.2

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=2ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping PC3 ke PC5

c:\>ping 192.168.30.3

```
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time=2ms TTL=126
Reply from 192.168.30.3: bytes=32 time=1ms TTL=126
Reply from 192.168.30.3: bytes=32 time=1ms TTL=126
Reply from 192.168.30.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Ping PC4 ke PC7

c:\>ping 192.168.40.3

```
C:\>ping 192.168.40.3

Pinging 192.168.40.3 with 32 bytes of data:

Reply from 192.168.40.3: bytes=32 time=3ms TTL=126
Reply from 192.168.40.3: bytes=32 time=1ms TTL=126
Reply from 192.168.40.3: bytes=32 time=1ms TTL=126
Reply from 192.168.40.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Ping PC6 ke PC1

c:\>ping 192.168.10.3

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=6ms TTL=124
Reply from 192.168.10.3: bytes=32 time=3ms TTL=124
Reply from 192.168.10.3: bytes=32 time=5ms TTL=124
Reply from 192.168.10.3: bytes=32 time=3ms TTL=124

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 4ms
```

Ping PC4 ke PC0

c:\>ping 192.168.10.2

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=5ms TTL=125
Reply from 192.168.10.2: bytes=32 time=3ms TTL=125
Reply from 192.168.10.2: bytes=32 time=4ms TTL=125
Reply from 192.168.10.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms
```

Ping PC2 ke PC7

c:\>ping 192.168.40.3

```
C:\>ping 192.168.40.3

Pinging 192.168.40.3 with 32 bytes of data:

Reply from 192.168.40.3: bytes=32 time=4ms TTL=125
Reply from 192.168.40.3: bytes=32 time=6ms TTL=125
Reply from 192.168.40.3: bytes=32 time=2ms TTL=125
Reply from 192.168.40.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms
```